

CH. MONESH ARCHAN

B.Tech CSE (Data Science) Student · VIT Vellore

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PROJECTS

AI-Powered Medical Drone System

AEROMED-AI

React.js, Vite, JavaScript, Python, FastAPI, REST APIs, Leaflet, OpenStreetMap, Recharts, ESP32, Arduino C++, DS18B20

- Developed a React-based command-center dashboard for managing emergency medical drone operations and real-time mission information.
- Built an AI-driven workflow for prioritizing emergency requests and assigning suitable drones based on operational requirements.
- Implemented route visualization, no-fly-zone handling, and drone fleet/mission simulation using interactive map components.
- Integrated ESP32 and DS18B20 sensors for real-time monitoring of temperature-sensitive medical supplies.

AI Resume Analyzer

Python, FastAPI, React.js, NLP, spaCy, PyPDF2, JavaScript, REST APIs

- Developed a web-based resume analysis system that extracts candidate skills and evaluates resumes against job descriptions.
- Implemented resume parsing and skill extraction using Python, PyPDF2, and NLP techniques.
- Built ATS compatibility scoring and job-description matching functionality to identify relevant skills and improvement areas.
- Integrated the React.js frontend with a FastAPI backend through REST APIs for resume processing and analysis.

IoT Assistive System

Smart Blind Stick

ESP32, HC-SR04, NEO-6M GPS, Blynk IoT, Arduino IDE, Embedded C/C++

- Designed and programmed an ESP32-based assistive device for obstacle detection and emergency location tracking.
- Integrated ultrasonic sensors, GPS, SOS button, buzzer, vibration motor, and LDR for obstacle awareness and emergency assistance.
- Developed embedded firmware to process sensor data and trigger vibration and audio alerts based on detected obstacles.
- Built a Blynk IoT dashboard for remote sensor monitoring, GPS coordinates, and SOS status.

EXPERIENCE

Summer Intern

05/2022 – 08/2022

Adela Electricals Pvt. Ltd.

Electrical Systems & Manufacturing Intern

- Observed electrical panel manufacturing, wiring processes, and industrial electrical components.
- Gained practical exposure to electrical panel assembly, component identification, and wiring practices.
- Developed a product during the internship and presented its working and features to industry professionals.

Data Analytics Virtual Experience

Deloitte

- Analyzed industrial telemetry data to identify equipment-related patterns and potential downtime issues.
- Worked with data analysis and visualization concepts to communicate findings through dashboards.
- Applied data-driven reasoning to identify operational insights from machine telemetry.

EDUCATION

Vellore Institute of Technology (VIT),

Vellore

Expected 2029

B.Tech in Computer Science and Engineering (Data Science)

Narayana Junior College

98.1%

Class XII

ZPHS Mettu

98%

Class X

TECHNICAL SKILLS

Programming: C, C++, Java, Python, JavaScript, SQL

Web Development: HTML, CSS, React.js, Vite, FastAPI, REST APIs, Tailwind CSS

Data Science & AI: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, NLP, spaCy

Databases: MySQL, MongoDB

Tools & Platforms: Git, GitHub, VS Code, XAMPP, Linux, Cisco Packet Tracer, Arduino IDE

IoT & Embedded: ESP32, Blynk IoT, Embedded C/C++

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS & ACHIEVEMENTS

- Deloitte Data Analytics Virtual Experience Certificate
- Startup Street X Participation Certificate
- InnoHack Hackathon — Participant
- Participant in 2 Hackathons
- Gravitas — Workshop Participant

ACTIVITIES

RoboVITics, VIT Vellore

Club Member

FIND ME ONLINE

GitHub

github.com/monesharchan07

Portfolio

monesharchan07.github.io/portfolio/